

The metabolomics data acquisition for feline plasma and urine samples was performed at the West Coast Metabolomics Center (Davis, California), a National Institute of Health regional resource laboratory on May 2017 for the testing data set samples and March 2016 for the training data set samples. An in-depth description of the methods utilized have been previously published and are summarized below (Fiehn and Kind 2007, Fiehn 2017 and Rivera 2019).

Extraction, clean-up and derivatization

A 30 μ L aliquot of urine sample was added to a 1.0 mL cooled extraction solution composed of acetonitrile and isopropanol (JT Baker Center Valley, PA, USA) and 18.2 M Ω water (MilliQ) at a 3:3:2 v/v/v ratio in a 1.5 mL polypropylene tube. Before adding the extraction mixture to the study samples, oxygen was removed by sparging nitrogen for 5 minutes. Internal standards with isotope-labeled metabolite surrogates were added to the study samples. The study samples were mixed by vortexing for 10 s and then placed in a chilled (4 $^{\circ}$ C) shaker for 5 mins. Samples were then centrifuged at 14,000 rcf for 2 minutes to pellet insoluble material. A 450 μ L aliquot of the supernatant was removed, placed in a fresh 1.5 mL polypropylene tube and evaporated using a cold trap concentrator. The dried samples were re-suspended with 450 μ L of 50% acetonitrile and centrifuged at 14,000 rcf for 2 minutes. The supernatant was transferred to a new tube and desiccated using a cold trap concentrator. The dried samples were derivatized by adding 10 μ L of 40 mg/mL of methoxyamine hydrochloride (Aldrich, St. Louis, MO, USA) and shaken at maximum speed at room temp 30 $^{\circ}$ C for 1.5 h. A 10 μ L aliquot of fatty acid methyl esters (FAME, recipe provided in Fiehn 2017) was added to 1 mL of N-methyl-N-(trimethylsilyl)-trifluoroacetamide (MSTFA; Aldrich, St. Louis, MO, USA) and vortexed for 10 s. Finally, 91 μ L of the MSTFA + FAME mixture was added to each sample at shaken at 37 $^{\circ}$ C at maximum speed for 30 minutes. The final mixture was transferred to glass vials containing micro-inserts and immediately sealed. A reagent and derivatization blank was used per batch of 50 sample extractions as negative control. A reagent blank uses all reagents, but does not use extraction tubes or extraction solvents.

Chromatography and mass spectrometry data acquisition

Samples were separated using an RTX-5Sil MS analytical column made of 0.25 μ m thick 95% dimethyl/5% diphenylpolysiloxane film which is 30 m long with a 0.25 mm internal diameter (Restek corporation, Bellefonte, PA USA). This analytical column is surrounded and protected by a 10 m long guard column. The mobile phase consisted of pure helium with a flow-rate of 1 mL/min and a column temperature of 50-330 $^{\circ}$ C. The sample injection volume was 0.5 μ L at 50 $^{\circ}$ C ramped to 250 $^{\circ}$ C by 12 $^{\circ}$ C/s. The oven was held at 50 $^{\circ}$ C for 1 minute and then ramped to 330 $^{\circ}$ C and held for 5 minutes. The ramp speed for the oven is 20 $^{\circ}$ C/min. An Agilent 6890 gas chromatograph with a Leco Pegasus IV gas chromatography, time-of-flight mass spectrophotometer (GC-TOF-MS) controlled with Leco ChromaTOF software version 2.32 were used for metabolite data acquisition. The mass spectrophotometer parameters included unit mass resolution at 17 spectra/s from 80-500 Da at -70 eV ionization energy and 1800 V detector voltage with a 230 $^{\circ}$ C transfer line and a 250 $^{\circ}$ C ion source.

Data processing and metabolite identification

Raw data files were preprocessed using ChromaTOF version 2.32 with the following settings: no smoothing, 3 s peak width, baseline subtraction, automatic spectral deconvolution and peak detection at signal/noise levels of 5:1 throughout the chromatogram. Spectra which failed to meet at least one of these parameters were considered not identified for purposes of this study. All spectra identification was done using the BinBase database.

The BinBase algorithm (rtx5) used the following settings: validity of chromatogram (10^7 counts s $^{-1}$), unbiased retention index marker detection (MS similarity >800, validity of intensity

range for high m/z marker ions), retention index calculation by 5th order polynomial regression. Spectra were cut to 5% base peak abundance and matched to database entries from most to least abundant spectra using the following matching filters: Parameters used to identify spectra included retention index window ± 2000 U (around ± 2 s retention time deviation), unique ion and apex mass validation, mass spectrum similarity and a final isomer filter. Peak height was reported as quantification using the unique ion as default. A quantification report table was produced for all database entries that are positively detected in more than 10% of the samples of a study design class for unidentified metabolites. A subsequent post-processing module was employed to automatically replace missing values. Replaced values were labeled as 'low confidence' by color coding, and for each metabolite, the number of high-confidence peak detections was recorded as well as the ratio of the average height of replaced values to high-confidence peak detections. These ratios and numbers were used for manual curation of automatic report data sets to data sets released for submission. Peak heights (m/z value) at the specific retention index were reported. Peak heights instead of peak areas because peak heights are more precise for low abundant metabolites than peak areas, due to the larger influence of baseline determinations on areas compared to peak heights.

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